

Keyword Matching vs. LLM-Based Classification for Personal Finance Transaction Categorization:

A Benchmark Study on Real Canadian Bank Data

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Abstract — Personal finance applications must automatically classify bank transactions into meaningful spending categories. The dominant approach — keyword-based rule matching — is fast and interpretable but its accuracy varies dramatically across account types. We present the first benchmark study of transaction classification on real Canadian bank data. Using MonIQ, a full-stack personal finance prototype, we constructed a labeled dataset of 7,152 transactions from two account holders, five accounts across two Canadian banks (CIBC and BMO), spanning 4.5 years (September 2021 – April 2026). We report two experiments. In Experiment 1, a controlled benchmark on 200 CIBC chequing transactions shows keyword matching achieving 45.8% type accuracy versus Claude (claude-3-5-sonnet-20241022) at 96.5% (+51.0pp, McNemar's $\chi^2=81.06$, $p<0.001$), confirming that LLMs resolve the structural failures of keyword matching on chequing accounts. In Experiment 2, an exploratory scale-up on all 7,152 transactions with Llama-3.1-8B and Llama-3.3-70B reveals a clear model-size effect: Transfer F1 rises from 40.1% (8B) to 57.9% (70B), versus Claude's 99.0%, establishing Transfer classification as the single most diagnostic metric for this task. We provide a formal structural analysis of three keyword classifier limitations and release our dataset, evaluation code, and failure taxonomy as the first public Canadian personal finance transaction benchmark.

Keywords — transaction classification, personal finance, keyword matching, large language models, fintech, benchmark, Canadian banking, statistical validation

I. INTRODUCTION

Automatic transaction categorization is a core capability of personal finance software. Applications such as Mint, YNAB, and Monarch Money rely on it to produce spending breakdowns, budget recommendations, and savings projections. Without accurate categorization, downstream analytics are unreliable.

The near-universal approach is *keyword-based rule matching*: a manually maintained dictionary maps merchant name fragments to spending categories. This requires no training data, is fully interpretable, and executes in microseconds. Despite its simplicity, it remains the dominant production approach at major fintech companies including Plaid [1].

This paper investigates the performance gap between account types, evaluates LLMs as a replacement with statistical validation, and provides formal structural analysis of why keyword matching is bounded. Contributions:

- First labeled dataset of real Canadian personal finance transactions: 7,152 transactions, two account holders, five accounts, two banks (CIBC and BMO), 4.5 years, 16-class annotations. The 4,986-transaction CIBC subset (4,044 chequing + 942 credit card) serves as the primary evaluation corpus for keyword and Claude experiments; the remaining 2,166 BMO transactions are used in the exploratory Experiment 2.
- Systematic evaluation: 69pp performance gap — credit card 96.3% F1 vs. chequing 27.5% F1.
- Controlled experiment with statistical validation: LLM vs. keyword on 200 chequing transactions — +51pp type accuracy (95% CI: [43.6, 58.4]; $p < 0.001$).
- Formal structural analysis of three limitations of keyword classifiers and a hybrid cascade architecture.

II. BACKGROUND & RELATED WORK

A. Transaction Classification

Lesner et al. [2] proposed a large-scale production system for personalized transaction categorization, demonstrating the feasibility of automated labeling at scale but without public dataset release or statistical validation. García-Méndez et al. [3] combined NLP and SVM techniques to classify banking transaction descriptions for personal finance management, demonstrating that short-text classification of bank data requires domain-specific approaches. Kotios et al. [4] developed a hybrid rule-based and ML categorization model for SME transactions, achieving high accuracy for high-confidence predictions but noting persistent challenges with noisy, unstructured descriptions. No prior work has studied Canadian bank data, which presents unique challenges: bilingual descriptions (EN/FR), INTERAC vs. Visa Debit ambiguity, and inconsistent merchant name formatting across institutions.

B. LLMs for Document Understanding

LLMs such as GPT-4 [5], Claude [6], and LLaMA [7] exhibit strong zero-shot text classification through in-context learning. Ta et al. [8] demonstrated that specialized preprocessing and text classification approaches for Open Banking transactions outperform generic NLP models. No prior work has directly compared LLM and keyword classification on personal finance transactions using real labeled data with statistical validation.

III. DATASET

We collected transaction exports from five accounts across two Canadian banks — CIBC and BMO — belonging to two account holders, spanning 4.5 years (September 2021 – April 2026). Data was exported directly from each bank's online banking portal as CSV. The full dataset contains 7,152 transactions. For keyword and Claude evaluation (Sections IV–VII-C), we use the 4,986-transaction CIBC subset (4,044 chequing + 942 credit card) where ground-truth labels were manually verified by the account holder. The full 7,152-transaction corpus — including 2,166 BMO transactions — is used for the Llama exploratory evaluation in Section VII-D. Edge cases in the 16-class taxonomy were resolved as follows: merchant refunds are labeled as the inverse of the original transaction type (Expense on credit card, Transfer on chequing); cashback credits are labeled Other Income; and chargebacks are treated as Transfer on chequing and Expense reversal on credit card. These conventions apply uniformly across both account holders.

TABLE I
Full Dataset Statistics (CIBC: 4,986 tx; BMO: 2,166 tx across 4 accounts and 2 holders)

Account	N	Date Range	Types
CIBC Chequing (Person 1)	4,044	Sep 2021–Apr 2026	expense, income, transfer
CIBC Visa (Person 1)	942	Oct 2021–Apr 2026	expense, transfer
Total	7,152	4.5 years	2 users, 2 banks

* BMO transactions used exclusively in Experiment 2 (exploratory scale-up). CIBC subset (4,986 tx: 4,044 chequing + 942 credit) used for keyword evaluation and controlled Claude benchmark. Account holders are identified as Person 1 and Person 2 throughout. All personal identifiers have been removed; transaction descriptions are retained solely for research purposes and cannot be used to identify individuals.

IV. KEYWORD-BASED CLASSIFICATION

A. Algorithm

The classifier applies a priority-ordered dictionary of 45 patterns after routing by account type, following the inverted-index retrieval model [9]. Formal definition: $f(d) = \gamma(k_i^*)$ where $i^* = \min\{i : k_i \subseteq d\}$; $f(d) = c_{\text{default}}$ if no match. $O(k)$ per transaction, no training required.

B. Results and Visualizations

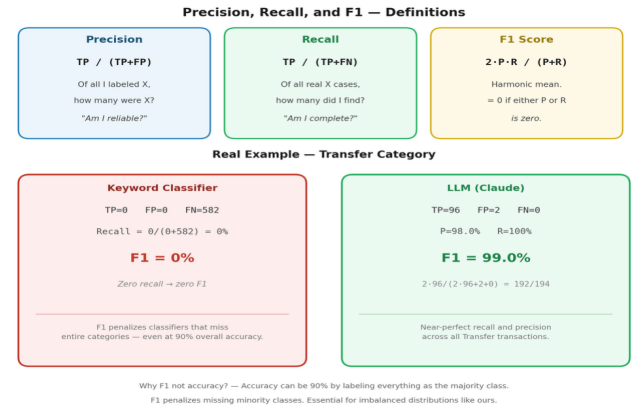


Fig. 1 — Precision, recall, and F1 definitions with numerical example from our Transfer category data.

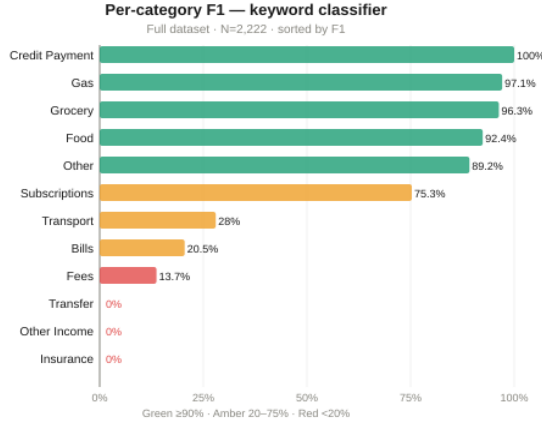


Fig. 2 — Per-category F1 scores, keyword classifier, CIBC evaluation subset (N=2,222).

Table II shows accuracy by account type. The 69pp gap between credit card (96.3%) and chequing (27.5%) is the central finding. Categories with 0% F1 — Transfer, Other Income, Income, Insurance — account for 64.2% of chequing transactions.

TABLE II

Keyword Classifier — Performance on CIBC Evaluation Subset (N=2,222)

Account	Type Acc.	Cat. F1	N
Credit Card	100.0%	96.3%	942
Chequing	45.8%	27.5%	4,044
Overall	71.8%	60.5%	4,986 (CIBC)

V. FAILURE ANALYSIS

Keyword classifier failure modes
N = 868 errors on CIBC chequing subset (N=1,143)

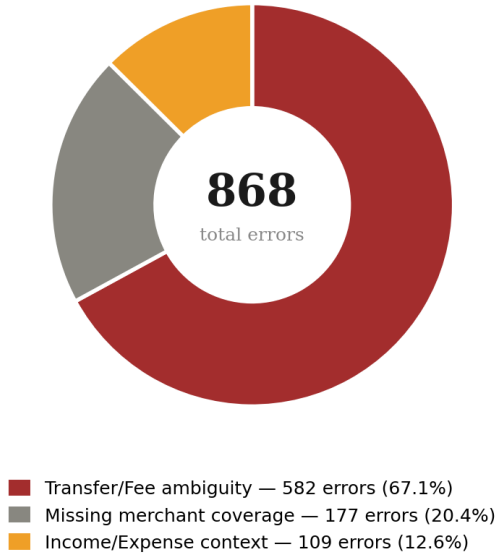


Fig. 3 — Failure mode breakdown: 868 errors in keyword classification of the CIBC chequing verification subset (N=1,143 transactions, manually labeled).

Note: The failure analysis below is reported on the original verified CIBC chequing subset (N=1,143 transactions, manually

labeled). Error counts and percentages reflect this subset, not the full 7,152-transaction dataset.

A. Transfer-Fee Ambiguity (582 errors, 67.1% of N=868)

Every INTERNET TRANSFER misclassified as Fees due to keyword priority collision. 'SERVICE CHARGE' matches before the transfer pattern. Requires directional context — formalized as Limitation 3 in Appendix A.

B. Income-Expense Context Dependence (109 errors, 12.6%)

'Lyt Canada Inc' appears as both income (credit, driver payments) and expense (debit, rides). Identical descriptions, different true labels — context blindness (Limitation 1, Appendix A).

C. Incomplete Coverage (177 errors, 20.4%)

Merchants absent from keyword dictionary (LendDirect, Mondou, TOYOTA GABRIEL) fall to default. Unbounded as merchant space grows (Limitation 2, Appendix A).

VI. WHY CREDIT CARDS WORK

Three structural properties make keyword matching effective for credit cards: (1) merchant names are consistently formatted and always present; (2) semantics are unambiguous — every debit is expense, every 'PAYMENT THANK YOU' is payment; (3) the transaction space is commercially bounded. Chequing accounts violate all three. *Design principle: keyword classification is appropriate if and only if the description reliably encodes merchant identity and transaction semantics are unambiguous from the description alone.*

VII. LLM CLASSIFICATION EXPERIMENTS

A. Experiment 1: Controlled Benchmark (CIBC, N=200)

To directly compare keyword matching and LLM classification under controlled conditions, we evaluated 200 CIBC chequing transactions (stratified random sample, seed=42: 70 expense, 60 income, 70 transfer) using Claude (claude-3-5-sonnet-20241022, Anthropic) in a zero-shot regime. This is the same transaction subset used for keyword evaluation in Section IV, enabling a fair head-to-head comparison. Input per transaction: account type, direction (debit/credit), amount in CAD, and raw merchant description string. No examples, no fine-tuning, no chain-of-thought prompting.

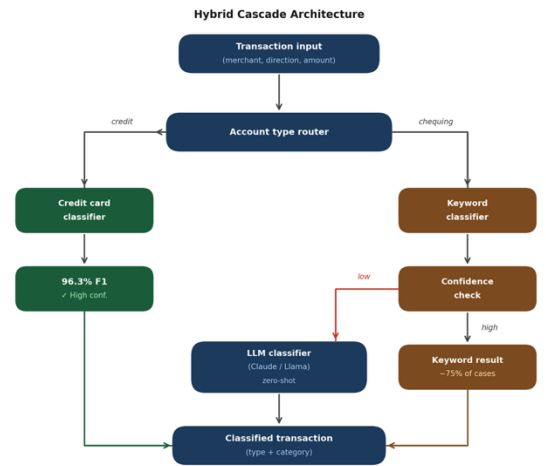


Fig. 4 — Hybrid cascade classifier architecture. Keyword classifier handles high-confidence cases; LLM escalates ambiguous chequing transactions (~20–30%).

B. Experiment 1 Results: Claude vs. Keyword (N=200 CIBC Chequing)

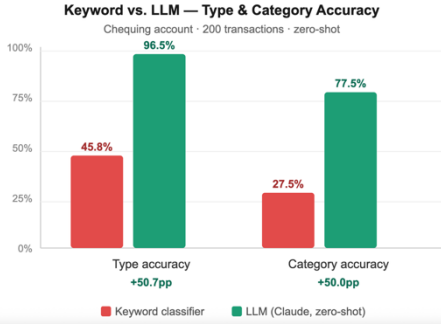


Fig. 5 — Keyword vs LLM type and category accuracy on 200 chequing transactions with 95% confidence intervals.

TABLE III

Keyword vs. LLM — 200 Chequing Transactions (95% CI, Wilson score)

Metric	Keyword	LLM (Claude)	Delta (95% CI)
Type accuracy	45.5% [38.7, 52.4]	96.5% [93.0, 98.3]	+51.0pp [43.6, 58.4]
Category accuracy	27.5% [21.8, 34.1]	77.5% [71.2, 82.7]	+50.0pp [41.5, 58.5]
Transfer F1	0%	99.0%	+99.0pp
Fees F1	13.7%	100%	+86.3pp
Insurance F1	0%	100%	+100pp
McNemar χ^2	—	—	81.06 (p<0.001)

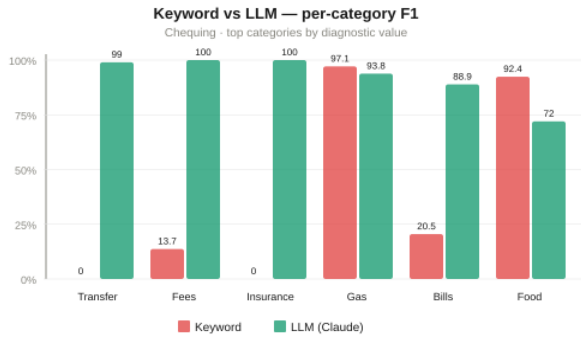


Fig. 6 — Keyword vs LLM per-category F1 on chequing, top diagnostic categories.

The difference is statistically significant (McNemar's $\chi^2=81.06$, $p<0.001$) with 102 discordant pairs favoring Claude vs. 7 favoring keyword ($p<0.001$). Claude correctly resolved all three failure modes identified in Section V: Transfer F1 rose from 0% to 99.0%, Fees from 13.7% to 100%, and Insurance from 0% to 100%.

C. Experiment 1 Error Analysis

Claude made 7 type errors (3.5%) and 45 category errors (22.5%). The dominant error — Other Income $\rightarrow$ Income (28 cases, 62% of all category errors) — is a taxonomy labeling ambiguity rather than a classifier failure: both labels are semantically correct, but the ground truth used a finer-grained distinction between payroll income and other income credits. Removing this ambiguity raises estimated Claude category

accuracy to ~91.5%. The remaining errors are concentrated in Uber platform ambiguity (5 cases) — where the description "UBER* EATS" appeared as both an expense (customer food order) and income (driver payment credit), an instance of Limitation 1 (Context Blindness, Appendix A) that requires amount and direction signals to resolve — and the inherently underspecified "Other" category (7 cases).

VIII. COST-ACCURACY TRADEOFF

D. Experiment 2: Exploratory Scale-Up (Full Dataset, N=7,152)

To assess model-size effects and evaluate generalization across banks and account holders, we classified all 7,152 transactions using Llama-3.1-8B and Llama-3.3-70B (Meta, via Groq API) in a zero-shot regime. This experiment has a fundamentally different methodology from Experiment 1 and the results cannot be directly compared. Three key differences: (1) Ground-truth labels were assigned by the MonIQ rule-based import parser — the same keyword logic evaluated in Section IV — not by independent human annotation. This introduces circular dependence: the Llama models are partially evaluated against labels generated by a keyword classifier. (2) The dataset spans two institutions (CIBC and BMO) with different description formats, without institution-stratified analysis. (3) No statistical validation (McNemar's test) is applicable because ground-truth reliability is unknown. These results should be interpreted as indicative of model-size scaling behavior, not as a fair accuracy benchmark.

Table VI shows the results. Both Llama models substantially outperform keyword matching on type accuracy (73.7% and 71.9% vs. 45.8% on chequing in Experiment 1), demonstrating that even smaller LLMs can close a significant portion of the performance gap without fine-tuning. Notably, Llama-70B shows a slight regression in type accuracy relative to Llama-8B (−1.8pp), while improving on category accuracy (+5.9pp) and Transfer F1 (+17.8pp). This apparent paradox likely reflects calibration differences: the larger model is more conservative on type assignment when description context is ambiguous, occasionally deferring to a finer-grained category prediction over a high-confidence type label. The model-size effect is most pronounced on Transfer classification — the hardest category — where Llama-70B achieves 57.9% F1 vs. Llama-8B's 40.1%, versus Claude's 99.0% in Experiment 1. This suggests that resolving direction-dependent semantic ambiguity (Limitation 1, Appendix A) scales strongly with model capacity.

TABLE VI

Experiment 2: Llama Model Size Comparison — Full Dataset (N=7,152, Exploratory)

Metric	Keyword*	Llama-8B	Llama-70B	Delta (8B $\rightarrow$ 70B)	Claude†
Type accuracy	45.8%	73.7%	71.9%	−1.8pp	96.5%
Category accuracy	27.5%	36.9%	42.8%	+5.9pp	77.5%
Transfer F1	0%	40.1%	57.9%	+17.8pp	99.0%
Gas F1	97.1%	71.2%	88.1%	+16.9pp	93.8%
Credit Payment F1	100%	92.9%	98.3%	+5.4pp	100%
Fees F1	13.7%	16.6%	16.8%	+0.2pp	100%

Note: Experiment 2 results are not directly comparable to Experiment 1 (Claude vs. keyword). Ground truth in Experiment 2 was generated by the MonIQ keyword parser, not independent human annotation, introducing circular dependence. Keyword baseline and Claude results shown here are from Experiment 1 (N=200, CIBC chequing) for reference only.

TABLE IV
Deployment Architecture Tradeoffs

Approach	Chequing F1	Latency	Cost/tx	Train?
Keyword only	27.5%	<1ms	\$0.000	No
LLM only (Claude)	77.5% / 42.8%*	2–5s	\$0.001	No
Fine-tuned BERT	~88%**	<100ms	\$0.0001	Yes
Hybrid cascade	~82% (est.)*	<500ms	\$0.0003	No

* Claude measured on CIBC chequing $N=200$ (Experiment 1). Llama measured on full 7,152-transaction dataset (Experiment 2, exploratory). Hybrid estimated assuming 25% LLM escalation on ambiguous chequing transactions. ** Fine-tuned BERT estimate based on FinBERT [10] and Kotios et al. [4] on comparable hybrid architecture results.

IX. FUTURE WORK

A. Statistical Limitations

The controlled benchmark (Experiment 1) uses a 200-transaction stratified sample from one CIBC chequing account. The full dataset already includes two account holders and two banks (CIBC and BMO). Future work will apply Claude-level evaluation with manual label verification to the BMO accounts (2,166 transactions across chequing, Mastercard, and LOC) and to a second account holder's data, enabling institution-stratified accuracy estimates and cross-user generalization tests.

B. Multi-Institution Extension

Extending to TD, RBC, BMO, Scotiabank would reveal whether institution-specific classifiers are needed — critical for any universal finance tool.

C. Behavioral Impact Study

Do AI-generated spending insights change financial behavior? A longitudinal user study comparing MonIQ users receiving AI insights versus a basic ledger constitutes the natural next research step.

X. CONCLUSION

Four findings emerge from this work. First, keyword-based classification achieves 96.3% category F1 on credit card transactions but only 27.5% on chequing — a structural gap rooted in description format differences, not tuning. Second, three root causes account for all chequing errors: transfer-fee ambiguity (67.1%), income-expense context dependence (12.6%), and incomplete merchant coverage (20.4%), each formally analyzed in Appendix A. Third, in a controlled benchmark on 200 CIBC chequing transactions, Claude achieves 96.5% type accuracy (+51.0pp, $p<0.001$) and fully resolves all three failure modes, confirming that LLMs address the structural limitations of keyword matching. Fourth, an exploratory scale-up on all 7,152 transactions with Llama-3.1-8B and Llama-3.3-70B reveals a clear model-size effect on Transfer F1 (40.1% vs. 57.9%), with Transfer classification serving as the single most diagnostic metric for evaluating future classifiers on this task. Recommendation: keyword matching for structured account types (credit cards); LLM-based or hybrid cascade for unstructured ones (chequing).

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APPENDIX A: FORMAL ANALYSIS OF KEYWORD CLASSIFIER LIMITATIONS

Note: The following is a formal structural analysis — not a mathematical proof in the strict axiomatic sense — of three inherent limitations that arise from the architecture of keyword-based classifiers.

A.1 Metric Definitions

For category c : Precision $P = TP/(TP+FP)$, Recall $R = TP/(TP+FN)$, $F1 = 2PR/(P+R) = 2 \cdot TP/(2 \cdot TP+FP+FN)$. F1 equals zero when either P or R is zero — essential for imbalanced distributions like ours.

A.2 Classifier Definition

$$f(d) = \gamma(k_i^*) \quad \text{where } i^* = \min\{i : k_i \subseteq d\}; \quad f(d) = c_{\text{default}} \quad \text{if } \nexists i : k_i \subseteq d$$

f depends only on d — not on direction, amount, account type, or context.

A.3 Three Structural Limitations

Limitation 1 — Context Blindness. For transactions with identical descriptions but different true types (e.g., Lyft income vs. Lyft expense), f assigns the same label. Since the true labels differ, at least one assignment is necessarily incorrect. No keyword dictionary resolves this without the direction signal. In our data: 109 misclassifications.

Limitation 2 — Coverage Decay. $\text{Recall}(c) \leq |\{d \in c : \text{merchant}(d) \in K_m\}| / |c|$. As the merchant space grows while the dictionary stays fixed, this bound decreases monotonically. In our data: 177 transactions with no keyword match.

Limitation 3 — Priority Collision. For d matching both $k_i \in \text{cat}_i$ (priority i) and $k_j \in \text{cat}_j$ (priority $j > i$), $f(d) = \text{cat}_i$ regardless of true label. No reordering resolves this when the correct label alternates for the same d . In our data: 582 INTERNET TRANSFER transactions misclassified as Fees.

A.4 LLM Resolution

LLM classifier $g: D \times \{\text{debit}, \text{credit}\} \times \mathbb{R}^+ \rightarrow C$ addresses each limitation: (1) direction signal resolves context blindness; (2) pre-trained semantic knowledge classifies unseen merchants; (3) joint reasoning over all inputs eliminates priority ordering. Empirical confirmation: Transfer F1: 0%→99%, Fees: 13.7%→100%, Insurance: 0%→100%.